

Lesson Plan — Electrostatics (charges, Coulomb's law, electric field, flux, Gauss's law)

Physics | Grade 11-12 | Intermediate | Electrostatics / Electric Field and Gauss's Law

Overview

Subject	Physics
Grade band	11-12
Topic	Electrostatics (charges, Coulomb's law, electric field, flux, Gauss's law)
Chapter	Electrostatics / Electric Field and Gauss's Law
Source document	1.4.1 Additivity of charges
Periods	6 x 40 min

Learning Objectives

- [apply] Apply Coulomb's law in vector form to calculate the electrostatic force on a point charge due to one other point charge given their position vectors and magnitudes.
- [apply] Compute the net electrostatic force on a charge due to multiple point charges by using the principle of superposition and vector addition.
- [analyze] Use Gauss's law to derive the magnitude of the electric field at distance r from (a) an infinitely long uniformly charged wire, (b) an infinite uniformly charged plane, and (c) outside and inside a uniformly charged thin spherical shell.
- [understand] Explain, using evidence from atomic-scale charge transfer, why rubbing two materials transfers electrons and does not create net charge.
- [analyze] Given a simple dipole (charges $+q$ and $-q$ separated by $2a$), derive the far-field ($r \gg a$) dependence of its electric field and express it in terms of the dipole moment p .
- [remember] Define surface, linear, and volume charge densities (σ , λ , ρ) from macroscopic charge elements and state their SI units.
- [understand] Describe the operational definition of the electric field at a point in space and state its SI unit as N/C.

Period-by-Period Plan

Period 1: Charge, Conductors, and Conservation

Concepts	Electrostatics, Conductors and insulators, Conservation of charge
Objectives	Explain, using evidence from atomic-scale charge transfer, why rubbing two materials transfers electrons and does not create net charge.
Time allocation	Activate prior knowledge (5 min), Introduce new material (10 min),

Teacher-led demonstration and mini-lab (12 min), Guided practice (6 min), Check for understanding (4 min), Consolidate and link forward (3 min)

Why this order

This opening period introduces what electric charge is at the atomic scale, how conductors and insulators differ, and the conservation of charge; those ideas set the physical vocabulary needed to state why charge is quantised and to derive Coulomb's law (magnitude and vector) in the following periods.

Period 2: Coulomb's Law and Charge Quantisation

Concepts	Quantisation of charge, Coulomb's law (magnitude form), Coulomb's law (vector form)
Objectives	Explain, using evidence from atomic-scale charge transfer, why rubbing two materials transfers electrons and does not create net charge.
Time allocation	Activate prior knowledge (5 min), Measured demonstration (8 min), Introduce Coulomb's law (magnitude) by derivation (10 min), Derive vector form (7 min), Guided practice (6 min), Formative check and consolidate (4 min)
Why this order	This period uses ideas from 'Charge, Conductors, and Conservation' (quantised free charge and conservation) to explain forces between point charges and prepares students to apply the principle of superposition and to define the electric field in the next period.

Period 3: Superposition and Electric Field Basics

Concepts	Principle of superposition, Electric field as characteristic of charge distribution, Superposition principle for electric field, Electric field lines representation, Electric flux through a surface element
Objectives	Explain, using evidence from atomic-scale charge transfer, why rubbing two materials transfers electrons and does not create net charge.
Time allocation	Activate prior knowledge (5 min), Introduce superposition for forces (10 min), Introduce electric field and superposition for (10 min), Demonstration and guided practice (8 min), Check and consolidate (7 min)
Why this order	This period builds directly on Coulomb's Law and Charge Quantisation by using the vector form of Coulomb's law to derive how forces and fields add from multiple point charges. Mastery of superposition and field description enables students to analyse dipole far-fields ($1/r^3$), apply Gauss's law, and calculate macroscopic electric fields in the next period.

Period 4: Coulomb Vectors, Dipole Far-Field, Gauss Review

Concepts	Far-field of a dipole ($1/r^3$ dependence), Gauss's law, Coulomb's law, Electric field
Objectives	Apply Coulomb's law in vector form to calculate the electrostatic force

on a point charge due to one other point charge given their position vectors and magnitudes.; Given a simple dipole (charges $+q$ and $-q$ separated by $2a$), derive the far-field ($r \gg a$) dependence of its electric field and express it in terms of the dipole moment p .; Describe the operational definition of the electric field at a point in space and state its SI unit as N/C.

Time allocation	Activate prior knowledge (5 min), Demonstration and guided example (10 min), Teacher-led derivation (12 min), Guided practice (8 min), Formative check (3 min), Consolidation and link forward (2 min)
Why this order	This period builds on the Superposition and Electric Field Basics material (superposition, field from distributions, and flux through an element) by applying Coulomb's law in vector form and deriving the dipole far-field; it prepares students for the following material on torque on a dipole in a uniform field, electric flux applications, and continuous charge distributions. Completing the vector force calculations and the $1/r^3$ dipole derivation makes it possible to use Gauss's law efficiently in later symmetry arguments and to compute dipole torques.

Period 5: Superposition, Charge Densities, Dipole Torque

Concepts	Superposition principle, Electric flux, Torque on a dipole in a uniform field, Continuous charge distributions (linear, surface, volume)
Objectives	Compute the net electrostatic force on a charge due to multiple point charges by using the principle of superposition and vector addition.; Define surface, linear, and volume charge densities (σ , λ , ρ) from macroscopic charge elements and state their SI units.
Time allocation	Activate prior knowledge (5 min), Introduce superposition (10 min), Demonstration & derivation (8 min), Introduce continuous charge densities (6 min), Guided practice (8 min), Check and consolidate (3 min)
Why this order	Builds directly on Coulomb Vectors, Dipole Far-Field, Gauss Review by using Coulomb's law vectorially and the dipole concept to derive net forces and torques; prepares students to compute fields of an infinitely long charged wire, an infinite charged plane, and a uniformly charged thin spherical shell.

Period 6: Gauss's Law: Fields of Wire, Plane, Sphere

Concepts	Field of an infinitely long uniformly charged wire, Field of an infinite uniformly charged plane, Field of a uniformly charged thin spherical shell
Objectives	Use Gauss's law to derive the magnitude of the electric field at distance r from (a) an infinitely long uniformly charged wire, (b) an infinite uniformly charged plane, and (c) outside and inside a uniformly charged thin spherical shell.
Time allocation	Activate prior knowledge (5 min), Derive field of infinite wire (10 min), Derive field of infinite plane (8 min), Derive thin spherical shell result (8

Why this order

min), Practice problems in pairs (5 min), Check and consolidate (4 min)
This period builds on 'Superposition, Charge Densities, Dipole Torque' by applying Gauss's law to common continuous charge distributions (linear λ , surface σ , total q). Successfully deriving these fields step-by-step makes it possible to solve boundary problems and estimate forces in extended-charge configurations; there is no following period since this closes the unit.

Concepts Taught

Concept	Importance	Summary
Electrostatics	core	The study of forces, fields and potentials arising from static charges.
Conductors and insulators	core	Conductors allow electricity to pass through them easily and have charges that are comparatively free to move; insulators offer high resistance and charges remain localized.
Conservation of charge	core	Within an isolated system the total electric charge is conserved; charge may be redistributed but the net charge remains constant.
Quantisation of charge	core	Free charges occur in integer multiples of a basic unit e (the electron charge); $q = n e$ with integer n .
Coulomb's law (magnitude form)	core	The electrostatic force between two point charges in vacuum is proportional to the product of charges and inversely proportional to the square of the separation.

Concept	Importance	Summary
Coulomb's law (vector form)	core	Vector form of Coulomb's law gives the force on one point charge due to another, including direction along the line joining them and sign-dependent attraction/repulsion.
Principle of superposition	core	The total electrostatic force on a charge due to several other charges is the vector sum of the individual forces from each charge, each given by Coulomb's law.
Electric field	core	An electric field is produced by a charge and exerts a force on a test charge placed in the field; for a point charge Q at the origin, $E(r) = (1/4\pi\epsilon_0) Q \hat{r} / r^2$.
Electric field as characteristic of charge distribution	supporting	The electric field is a vector field defined at every point in space that characterises the electrical environment of a system of charges and is independent of the test charge used to measure it.
Superposition principle for electric field	core	Net electric field due to a system of point charges equals the vector sum of the fields due to individual charges.

Concept	Importance	Summary
Electric field lines representation	supporting	Electric field lines are curves whose tangents give the direction of the field; density of lines represents field strength; lines originate on positive charges and terminate on negative charges.
Electric flux through a surface element	supporting	Electric flux through an area element ΔS is defined as the dot product $E \cdot \Delta S = E \Delta S \cos\theta$, representing the number of field lines crossing the element.
Far-field of a dipole ($1/r^3$ dependence)	supporting	At distances $r \gg a$, the electric field of a dipole falls off as $1/r^3$ and can be expressed in terms of the dipole moment p : on axis $E = (1/4\pi\epsilon_0) (2p / r^3)$, on equatorial plane $E = (1/4\pi\epsilon_0) (- p / r^3)$.
Gauss's law	core	Electric flux through a closed surface equals the total charge enclosed divided by ϵ_0 .
Coulomb's law	core	The mutual electrostatic force between two point charges is proportional to the product of the charges and inversely proportional to the square of their separation.

Concept	Importance	Summary
Superposition principle	core	For an assembly of charges, the net force (or field) at a point is the vector sum of contributions from each charge; presence of other charges does not alter the pairwise forces.
Electric flux	supporting	Flux through a small area element equals the dot product of the electric field and the vector area element; for closed surfaces flux sums are used in Gauss's law.
Torque on a dipole in a uniform field	supporting	A permanent dipole p in a uniform electric field E experiences a torque $\tau = p \times E$ and no net force.
Continuous charge distributions (linear, surface, volume)	core	Macroscopic charge distributions are described by linear (λ), surface (σ), and volume (ρ) charge densities defined as charge per unit length, area, and volume respectively.
Field of an infinitely long uniformly charged wire	core	Electric field at distance r from an infinite straight wire with linear charge density λ is $E = \lambda/(2\pi \epsilon_0 r)$ radially.
Field of an infinite uniformly charged plane	core	Electric field near an infinite plane with surface charge density σ is $E = \sigma/(2\epsilon_0)$ directed normal to the plane (same magnitude on both sides).

Concept	Importance	Summary
Field of a uniformly charged thin spherical shell	core	Outside a thin spherical shell of total charge q and radius R , the field is as if q were at the centre ($E = q/(4\pi\epsilon_0 r^2)$); inside the shell the field is zero.

Formulae

Quantisation of charge

$$q = n e$$

where q = electric charge on a body (C); n = integer (could be positive, negative or zero); e = magnitude of charge of an electron (basic unit of charge) (C)

Coulomb's law (scalar)

$$F = (1 / (4 \pi \epsilon_0)) * (q_1 q_2 / r^2)$$

where F = magnitude of electrostatic force between two point charges (N); q_1, q_2 = point charges (C); r = distance between the charges (m); ϵ_0 = permittivity of free space ($C^2 N^{-1} m^{-2}$)

Coulomb's law (vector)

$$F_{12} = (1 / (4 \pi \epsilon_0)) * (q_1 q_2 / r_{12}^2) * \hat{r}_{21}$$

where F_{12} = force on charge 1 due to charge 2 (vector) (N); $\hat{r}_{21}$ = vector from charge 2 to charge 1 (m)

Electric field of a point charge

$$E(r) = (1 / (4 \pi \epsilon_0)) * (Q / r^2) * \hat{r}$$

where $E(r)$ = electric field at position r ($N C^{-1}$ or $V m^{-1}$); Q = source point charge (C); r = distance from the source charge to field point (m)

Coulomb-like point-force expression (as presented)

$$F(r) = (1/4 \pi \epsilon_0) (Q q / r^2) \hat{r}$$

where Q, q = source and test charges (C); r = distance between charges (m); ϵ_0 = permittivity of free space ($C^2 N^{-1} m^{-2}$)

Force on test charge from field

$$F(r) = q E(r)$$

where F = force on test charge (N); q = test charge (C); $E(r)$ = electric field at position r ($N C^{-1}$)

Electric flux element

$$\Delta \Phi = E \cdot \Delta S = E \Delta S \cos \theta$$

where E = electric field ($N C^{-1}$); ΔS = area element vector (magnitude ΔS and direction normal) (m^2); θ = angle between E and the area normal (radians)

Dipole far-field (axis)

$$E(\text{on axis, } r \gg a) = (1/4 \pi \epsilon_0) (2 p / r^3) \text{ along } p$$

where p = dipole moment vector (C m); r = distance from dipole centre (m); ϵ_0 = permittivity of free space ($C^2 N^{-1} m^{-2}$)

Dipole far-field (equatorial)

E (on equatorial plane, $r \gg a$) = $-(1/4\pi\epsilon_0)(p/r^3)$

where p = dipole moment vector (C m); r = distance from dipole centre (m)

Coulomb's law (scalar/vector form)

F on 1 due to 2 = $(1/4\pi\epsilon_0) * (q_1 q_2 / r^2)$ directed along $\hat{r}$ from 1 to 2.

where q_1, q_2 = point charges (C); r = distance between charges (m); $\hat{r}_{12}$ = unit vector from charge 1 to charge 2; ϵ_0 = permittivity of free space (F m⁻¹)

Gauss's law (integral form, closed surface)

Total electric flux through closed surface S equals enclosed charge divided by ϵ_0 .

where E = electric field (N C⁻¹ (or V m⁻¹)); dS = vector area element (outward normal) (m²); q_{enc} = total charge enclosed by S (C); ϵ_0 = permittivity of free space (F m⁻¹)

Field of an infinite line charge

E at distance r from an infinite straight wire with linear charge density λ is $\lambda/(2\pi\epsilon_0 r)$ directed radially.

where λ = linear charge density (C m⁻¹); r = radial distance from wire (m); $\hat{n}$ = radial unit vector

Field of an infinite plane sheet

E near an infinite plane with surface charge density σ is $\sigma/(2\epsilon_0)$ normal to the plane.

where σ = surface charge density (C m⁻²); ϵ_0 = permittivity of free space (F m⁻¹); $\hat{n}$ = unit normal vector

Field of a uniformly charged thin spherical shell

Inside a uniformly charged thin spherical shell the field is zero; outside it is $q/(4\pi\epsilon_0 r^2)$ radially.

where q = total charge on shell (C); R = radius of shell (m); r = distance from centre (m)

Torque on electric dipole

Torque on a dipole in a uniform field equals p cross E .

where p = electric dipole moment (C m); E = electric field (N C⁻¹)